

Unit 3: Data Visualization with Power BI

3.1 Power BI Overview and Components

Power BI is a business analytics solution that lets you visualize your data and share insights across your organization.

The "Power" Components:

3.1.1. Power BI Desktop (The Authoring Tool)

Power BI Desktop is the foundational Windows application where the report lifecycle begins. It is primarily used by data analysts for **data discovery, modeling, and report creation**. In an exam, emphasize that this is where the "heavy lifting" occurs—including establishing table relationships and writing complex DAX formulas. Because it works locally, it allows for offline development, saving all configurations into a `.pbix` file before it is published to the cloud.

3.1.2. Power BI Service (The Cloud Hub)

The Power BI Service is a cloud-based **SaaS (Software-as-a-Service)** platform used for sharing and collaboration. Once a report is "Published" from the Desktop, it resides here. It is the only component where you can create **Dashboards** (consolidated views from multiple reports). It also handles the administrative side of BI, such as managing user permissions, creating workspaces for team collaboration, and scheduling automatic data refreshes.

3.1.3. Power BI Mobile (The Consumption App)

Power BI Mobile provides native apps for iOS and Android, ensuring that decision-makers can access insights on the go. These apps are optimized for touch and offer specialized features like **Push Notifications** for data alerts and **QR code scanning** to quickly pull up specific reports. In a school setting, an administrator might use this to monitor real-time attendance or exam results directly from their smartphone.

3.1.4. Power Query (The ETL Engine)

Power Query is the dedicated **ETL (Extract, Transform, Load)** engine used for data preparation. Often called the "data laundry," it allows users to clean messy data by removing duplicates, splitting columns, or changing data types without writing code. A key technical point for your exam: Power Query records every transformation step in the **M Language**, creating a repeatable script that runs automatically whenever the data is updated.

3.1.5. Power Pivot (The Modeling Engine)

Power Pivot is the in-memory data modeling engine responsible for creating relationships and advanced calculations. While Power Query cleans the data, Power Pivot makes it "intelligent" by linking tables into a **Star Schema**. It utilizes **DAX (Data Analysis Expressions)** to create measures (like Year-over-Year growth), enabling high-performance analytics on millions of rows of data within seconds.

3.1.6. Power BI Gateway (The Secure Bridge)

The Power BI Gateway acts as a secure "tunnel" that connects the cloud-based Power BI Service to **on-premises** data sources (like a local SQL server in a university basement). Since the cloud cannot directly access private local networks, the Gateway facilitates a secure data transfer. This ensures that the dashboards in the cloud stay updated with fresh local data without requiring manual uploads.

3.2 Connecting to Data Sources

Power BI can connect to virtually *any* data source.

Connection Modes:

1. **Import:** Data is uploaded into Power BI's memory. It's the fastest mode but has a 1GB limit (per dataset) for Pro users.
2. **DirectQuery:** Power BI queries the source directly in real-time. Best for massive Big Data volumes where importing isn't possible.

Common Sources:

- **Files:** Excel, CSV, JSON, XML, Folder.
- **Databases:** SQL Server, Oracle, MySQL, PostgreSQL.
- **Azure/Big Data:** Azure SQL, **Spark**, **Hive**, Google BigQuery.
- **Online Services:** Google Analytics, Salesforce, SharePoint.

3.3 Data Transformation and Modeling

Before you build a chart, you must "shape" the data in the **Power Query Editor**.

Transformation Steps:

3.3.1. Cleaning (Data Sanitization)

The cleaning phase is the first and most critical step in the transformation process, as it ensures the "Veracity" (accuracy) of the data. This involves identifying and removing **null values** (missing data), **duplicates** (repeated records), and **errors** that might have occurred during data entry or extraction. For example, in an education database, a student might have been accidentally registered twice; the cleaning process detects this redundancy and keeps only a single, unique record. By removing "noise" and correcting inconsistencies, this step prevents skewed results and ensures that any visualizations built afterward are based on trustworthy, high-quality information.

3.3.2. Shaping (Data Restructuring)

Shaping refers to modifying the structure of the data to make it more suitable for analysis. This includes **splitting columns**, such as breaking a "Full Name" column into "First Name" and "Last Name" to allow for better sorting or filtering. It also involves **changing data types**, such as converting a column of numbers that was imported as "Text" back into a "Whole Number" or "Decimal" so that mathematical calculations can be performed. Another advanced shaping technique is **Unpivoting**, which transforms wide tables (with many columns for months or years) into long, narrow tables, making them much easier to visualize in charts and graphs.

3.3.3. Merging and Appending (Data Integration)

Merging and Appending are the two primary methods for combining data from multiple sources into a single, unified table. **Appending** is a vertical operation—it's like "stacking" tables on top of each other. For example, if you have separate attendance files for January, February, and March with the same columns, you append them to create one master attendance sheet. **Merging**, on the other hand, is a horizontal operation similar to a SQL Join. It links two different tables based on a common column (a "Key"), such as merging a "Student Grade" table with a "Student Profile" table using the "Student ID" to bring names and grades together into one view.

Modeling Principles (Star Schema):

1. Fact Tables (The "Numbers")

The **Fact Table** is the central table in a star schema containing quantitative, measurable data. It records events or observations, such as sales transactions or exam scores.

- **Focus:** "What" happened (e.g., 95 marks scored).
- **Structure:** High volume of rows but few columns, containing mostly numbers and ID keys.

2. Dimension Tables (The "Context")

Dimension Tables surround the fact table to provide descriptive attributes. They answer questions about the data, such as Who, Where, and When. For example, while a Fact table has a score, the Student Dimension has the student's name and age.

- **Focus:** "Who/Where/When" it happened (e.g., Student: Ram, Date: Jan 2026).
- **Structure:** Fewer rows but many columns with text descriptions.

3. Relationships (\$1:\ast\$)

Relationships are the links between tables using common ID columns. In a Star Schema, the gold standard is the **One-to-Many (\$1:\ast\$) relationship**.

- **Logic:** One record in a Dimension (e.g., one unique Student ID) can relate to many records in the Fact table (e.g., that student's multiple test results).
- **Flow:** Filters move from the "One" side (Dimension) to the "Many" side (Fact) to slice and dice the numbers.

3.4 Creating Visualizations & Dashboards

Visualizations are the charts, and a **Dashboard** is a single-page "snapshot" of your most important metrics.

Key Visual Types:

- **Comparison:** Bar and Column charts.
- **Trends:** Line and Area charts.
- **Relationship:** Scatter plots.
- **Distribution:** Histograms and Treemaps.
- **Interactive Features:**

- **Slicers:** On-screen buttons to filter data (e.g., filter by "Year").
- **Drill-down:** Clicking a "Year" bar to see "Months" inside it.
- **Cross-highlighting:** Clicking a segment in a Pie chart highlights related data in other charts on the same page.

3.5 Sharing and Collaborating

Once your report is ready in Power BI Desktop, you must **Publish** it.

3.5.1. Publishing (The First Step)

Publishing is the process of moving your completed report from the **Power BI Desktop** (local) to the **Power BI Service** (cloud). By clicking the "Publish" button, your .pbix file—which includes the data model and visuals—is uploaded to a specific workspace in the cloud. This is a mandatory step because reports living only on your desktop cannot be shared or viewed by others in real-time.

3.5.2. Workspaces (The Collaboration Room)

Workspaces are shared, cloud-based "folders" or containers used to group related reports and datasets. They serve as a collaborative environment where a team of developers can work together on the same content before it is finalized. For instance, an IT department might have a "University Analytics" workspace where multiple analysts build different reports for the administration.

3.5.3. Sharing and Embedding (Distribution)

Once a report is published, you can distribute it to stakeholders through various methods. **Sharing** typically involves sending a direct link or giving specific users access to the report. For more integrated collaboration, reports can be **embedded** directly into platforms like **Microsoft Teams**, **SharePoint**, or even **PowerPoint**, allowing managers to see live data without leaving their daily communication apps.

3.5.4. Row-Level Security / RLS (Data Privacy)

Row-Level Security (RLS) is a security feature that restricts data access based on the user's identity. Instead of creating ten different reports for ten different departments, you create **one** report and apply filters. For example, in a school system, "Teacher A" and "Teacher B" can both open the same report, but RLS ensures Teacher A only sees data for their own class, while Teacher B sees only theirs.